

CSC1003 Practice 6

Function

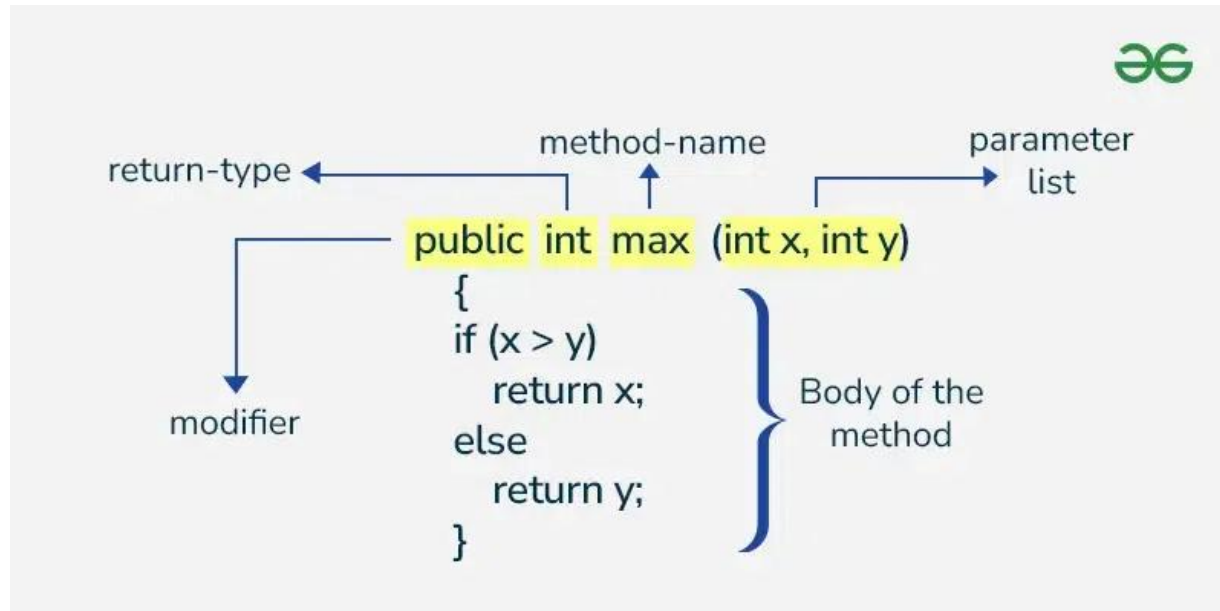
Brief Review

Why **Functions** (or the so-called **Methods** in Java) ?

- **Reusability**
 - Write once and use multiple times without repeating code.
- **Modularity**
 - Divide a program into separate methods to handle different tasks, more organized & easier to manage.
 - **Readability**: smaller & named methods are easier to understand.
 - **Maintainability**: easier to fix bugs or update the code when organized into separate methods.

Brief Review

How to Define **Functions** (or the so-called **Methods** in Java) ?



Acknowledgement: Adapted from GeeksforGeeks <<https://www.geeksforgeeks.org/java/methods-in-java/>>

Practice A: Triangle Perimeter Calculator

You're given three sets of coordinates that form a triangle, (x_i, y_i) with $i = 1, 2, 3$, and the length of each side can be calculated in the following manner:

$$d = \sqrt{(x_{i_1} - x_{i_2})^2 + (y_{i_1} - y_{i_2})^2}$$

Output the perimeter of the formed triangle, i.e. the sum of lengths of three sides.

Hint: Since you're calculating the three sides in the exact same way,

life will be much easier to write a method *calculateDistance* that takes in two sets of coordinates and calculates the distance between them.

Input:

```
1 2
8 7
5 -3
```

Expected Output:

```
25.445756013386024
```

Practice B: Swapping Elements

Write a method called *swapElements* that swaps two elements of an array.

- How can we do this?
- What parameters should it take (if any)?
- What should it return (if anything)?

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```
public void run() {  
    int[] array = new int[5];  
    ...  
    swapElements(array[0], array[1]);  
    ...  
}  
  
private void swapElements(int x, int y) {  
    int temp = x;  
    x = y;  
    y = temp;  
}
```

Practice B: Swapping Elements

Why it doesn't work?

Integers are primitives, so they are passed by value!

Their variable boxes store their actual values.

Therefore, changes to the parameter do NOT affect the original.

```
private void swapElements(int x, int y) {  
    int temp = x;  
    x = y;  
    y = temp;  
}
```

Practice B: Swapping Elements

After fix:

```
public void run() {  
    int[] array = new int[5];  
    ...  
    swapElements(array, 0, 1);  
    ...  
}  
  
private void swapElements(int[] arr, int pos1, int pos2) {  
    int temp = arr[pos1];  
    arr[pos1] = arr[pos2];  
    arr[pos2] = temp;  
}
```

Practice B: Swapping Elements

Why does this work?

```
private void swapElements(int[] arr, int pos1, int pos2) {  
    int temp = arr[pos1];  
    arr[pos1] = arr[pos2];  
    arr[pos2] = temp;  
}
```

- When you pass an array as a parameter to the method, the reference to that array is copied.
- Any changes in the content of the array through that reference will affect the original array.

*But changing the reference to point to a new array will NOT change the existing reference in the original method.

(Mid-Term) Q & A